

13.1 Vector Functions and Space Curves

In general, a function is a rule that assigns to each element in the domain an element in the range. A **vector-valued function**, or **vector function**, is simply a function whose domain is a set of real numbers and whose range is a set of vectors.

If $f(t)$, $g(t)$, and $h(t)$ are the components of the vector $\mathbf{r}(t)$, then f , g , and h are real-valued functions called the **component functions** of $\mathbf{r}$ and we can write

$$\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t) \mathbf{i} + g(t) \mathbf{j} + h(t) \mathbf{k}$$

We use the letter t to denote the independent variable because it represents time in most applications of vector functions.

EXAMPLE If $\mathbf{r}(t) = \langle t^3, \ln(3 - t), \sqrt{t} \rangle$

then the component functions are $f(t) = t^3$, $g(t) = \ln(3 - t)$, $h(t) = \sqrt{t}$

the domain of $\mathbf{r}$ consists of all values of t for which the expression for $\mathbf{r}(t)$ is defined. The expressions t^3 , $\ln(3 - t)$, and $\sqrt{t}$ are all defined when $3 - t > 0$ and $t \geq 0$. Therefore the domain of $\mathbf{r}$ is the interval $[0, 3)$.

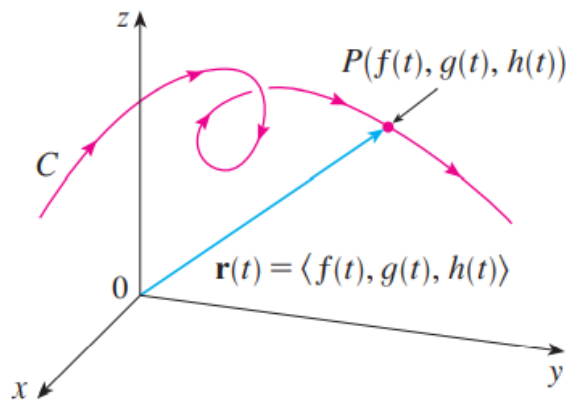
The **limit** of a vector function $\mathbf{r}$ is defined by taking the limits of its component functions as follows.

If $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$, then

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \left\langle \lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t) \right\rangle$$

provided the limits of the component functions exist.

EXAMPLE Find $\lim_{t \rightarrow 0} \mathbf{r}(t)$, where $\mathbf{r}(t) = (1 + t^3) \mathbf{i} + te^{-t} \mathbf{j} + \frac{\sin t}{t} \mathbf{k}$.



C is traced out by the tip of a moving position vector $\mathbf{r}(t)$.

Then the set C of all points (x, y, z) in space, where $x = f(t)$ $y = g(t)$ $z = h(t)$ and t varies throughout the interval I , is called a **space curve**.

The equations are called **parametric equations of C** and t is called a **parameter**.

EXAMPLE Describe the curve defined by the vector function

$$\mathbf{r}(t) = \langle 1 + t, 2 + 5t, -1 + 6t \rangle$$

EXAMPLE Sketch the curve whose vector equation is

$$\mathbf{r}(t) = \cos t \mathbf{i} + \sin t \mathbf{j} + t \mathbf{k}$$

[parametric plot \(cos t, sin t, t\) t = 0 to 30 - Wolfram|Alpha](#)

EXAMPLE Find a vector function that represents the curve of intersection of the cylinder $x^2 + y^2 = 1$ and the plane $y + z = 2$.

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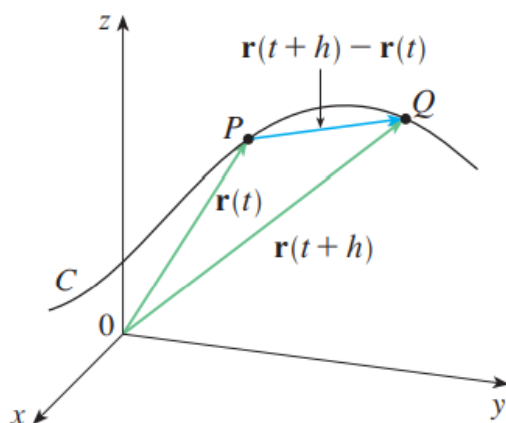
13.2 Derivatives and Integrals of Vector Functions

The **derivative** $\mathbf{r}'$ of a vector function $\mathbf{r}$ is defined in much the same way as for realvalued functions:

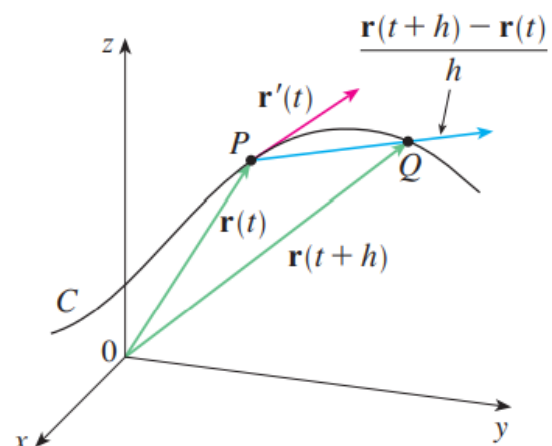
$$\frac{d\mathbf{r}}{dt} = \mathbf{r}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{r}(t + h) - \mathbf{r}(t)}{h}$$

The **tangent line** to C at P is defined to be the line through P parallel to the tangent vector $\mathbf{r}'(t)$.

the **unit tangent vector** is $\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$



(a) The secant vector $\overrightarrow{PQ}$



(b) The tangent vector $\mathbf{r}'(t)$

If $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t) \mathbf{i} + g(t) \mathbf{j} + h(t) \mathbf{k}$, where f , g , and h are differentiable functions, then

$$\mathbf{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle = f'(t) \mathbf{i} + g'(t) \mathbf{j} + h'(t) \mathbf{k} \quad \text{Fact}$$

EXAMPLE For the curve $\mathbf{r}(t) = \sqrt{t} \mathbf{i} + (2 - t) \mathbf{j}$,

(a) find $\mathbf{r}'(t)$ and sketch the position vector $\mathbf{r}(1)$ and the tangent vector $\mathbf{r}'(1)$.

(b) Find the unit tangent vector at the point where $t = 1$.

...

Suppose $\mathbf{u}$ and $\mathbf{v}$ are differentiable vector functions, c is a scalar, and f is a real-valued function. Then

$$1. \frac{d}{dt} [\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t)$$

$$2. \frac{d}{dt} [c\mathbf{u}(t)] = c\mathbf{u}'(t)$$

Differentiation Rules

$$3. \frac{d}{dt} [f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t)$$

$$4. \frac{d}{dt} [\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$$

$$5. \frac{d}{dt} [\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$$

$$6. \frac{d}{dt} [\mathbf{u}(f(t))] = f'(t)\mathbf{u}'(f(t)) \quad (\text{Chain Rule})$$

EXAMPLE Show that if $|\mathbf{r}(t)| = c$ (a constant), then $\mathbf{r}'(t)$ is orthogonal to $\mathbf{r}(t)$ for all t .

... ..

If $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$,

Integrals

$$\int_a^b \mathbf{r}(t) dt = \left(\int_a^b f(t) dt \right) \mathbf{i} + \left(\int_a^b g(t) dt \right) \mathbf{j} + \left(\int_a^b h(t) dt \right) \mathbf{k}$$

13.3 Arc Length and Curvature

the length of a plane curve with parametric equations

$$x = f(t), \quad y = g(t), \quad a \leq t \leq b,$$

$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2} dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

equivalently,

if the curve has the vector equation $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$, $a \leq t \leq b$,

or, the parametric equations $x = f(t), y = g(t), z = h(t)$,

then its length is

$$\begin{aligned} L &= \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2} dt \\ &= \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt \end{aligned}$$

into the more compact
form

$$L = \int_a^b |\mathbf{r}'(t)| dt$$

suppose that C is a curve given by a vector function

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k} \quad a \leq t \leq b$$

Now we define

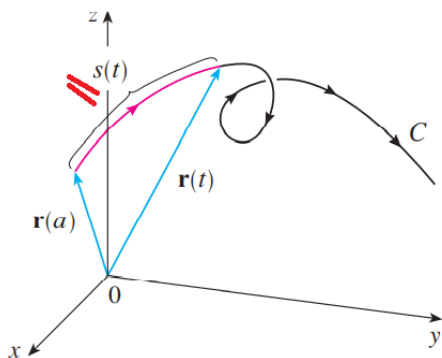
arc length function s by

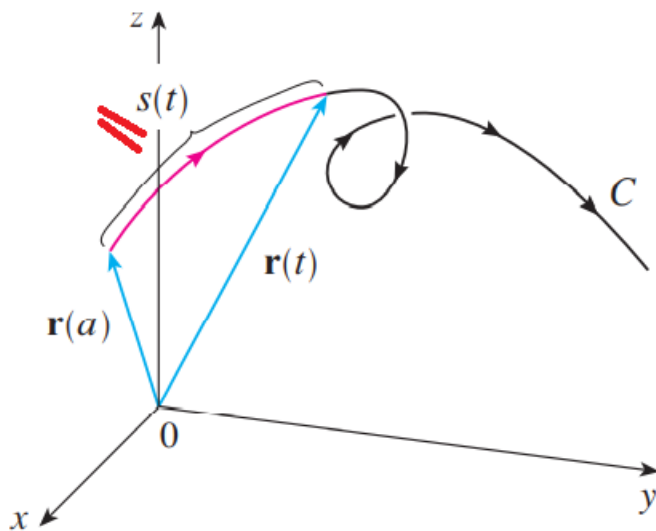
$$\underline{s(t)} = \int_a^t |\mathbf{r}'(u)| du = \int_a^t \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2 + \left(\frac{dz}{du}\right)^2} du$$

Thus $\underline{s(t)}$ is the length of the part of C between $\mathbf{r}(a)$ and $\mathbf{r}(t)$.

using Part 1 of the Fundamental Theorem of Calculus, we obtain

$$\frac{ds}{dt} = |\mathbf{r}'(t)|$$





suppose that C is a curve given by a vector function

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k} \quad a \leq t \leq b$$

Now we define

(6) **arc length function s** by

$$\underline{s(t)} = \int_a^t |\mathbf{r}'(u)| \, du = \int_a^t \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2 + \left(\frac{dz}{du}\right)^2} \, du$$

Thus $\underline{s(t)}$ is the length of the part of C between $\mathbf{r}(a)$ and $\mathbf{r}(t)$.

using Part 1 of the Fundamental Theorem of Calculus, we obtain

$$\frac{ds}{dt} = |\mathbf{r}'(t)|$$

It is often useful to **parametrize a curve with respect to arc length** because arc length arises naturally from the shape of the curve and does not depend on a particular coordinate system. If a curve $\mathbf{r}(t)$ is already given in terms of a parameter t and $s(t)$ is the arc length function given by Equation 6, then we may be able to solve for t as a function of s : $t = t(s)$. Then the curve can be reparametrized in terms of s by substituting for t : $\mathbf{r} = \mathbf{r}(t(s))$.

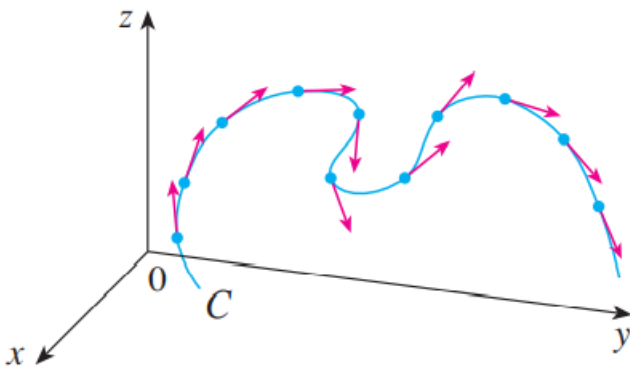
(6)

Curvature

If C is a smooth curve defined by the vector function $\mathbf{r}$, recall that the unit tangent vector $\mathbf{T}(t)$ is given by

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$$

The curvature of C at a given point is a measure of how quickly the curve changes direction at that point. = = ...
 = = ... the magnitude of the rate of change of the
 = = ... unit tangent vector with respect to arc length.



The **curvature** of a curve is where $\mathbf{T}$ is the unit tangent vector. $\kappa = \left| \frac{d\mathbf{T}}{ds} \right|$

Equivalently, $\frac{d\mathbf{T}}{dt} = \frac{d\mathbf{T}}{ds} \frac{ds}{dt}$ because and $\kappa = \left| \frac{d\mathbf{T}}{ds} \right| = \left| \frac{d\mathbf{T}/dt}{ds/dt} \right|$

we so have

$$\kappa(t) = \frac{|\mathbf{T}'(t)|}{|\mathbf{r}'(t)|}$$

The curvature of the curve given by the vector function $\mathbf{r}$ is

Fact

$$\kappa(t) = \frac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|^3}$$

EXAMPLE Find the curvature of the twisted cubic $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$ at a general point and at $(0, 0, 0)$ **ans.** $\kappa(0) = 2$.

For the special case of a plane curve with equation $y = f(x)$,

$\mathbf{r}(x) = x \mathbf{i} + f(x) \mathbf{j}$. Then

$$\kappa(x) = \frac{|f''(x)|}{[1 + (f'(x))^2]^{3/2}}$$

The Normal and Binormal Vectors

At a given point on a smooth space curve $\mathbf{r}(t)$, we can define the **unit normal vector** $\mathbf{N}(t)$ as (which is perpendicular to $\mathbf{T}(t)$)

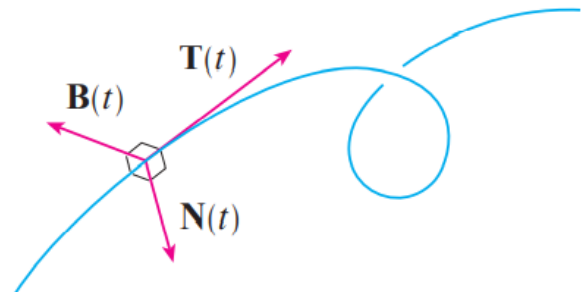
$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{|\mathbf{T}'(t)|}$$

Furthermore,

The vector $\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$ is called the **binormal vector**.

It is perpendicular to both $\mathbf{T}$ and $\mathbf{N}$ and is also a unit vector.

think of the normal vector as indicating the direction in which the curve is turning at each point.



In general, the vectors $\mathbf{T}$, $\mathbf{N}$, and $\mathbf{B}$, starting at the various points on a curve, form a set of orthogonal vectors, called the **TNB** frame, that moves along the curve as t varies. This **TNB** frame plays an important role in the branch of mathematics known as differential geometry and in its applications to the motion of spacecraft.

(see/ search “Frenet trihedron”)

https://en.wikipedia.org/wiki/Frenet%E2%80%93Serret_formulas#/media/File:Frenetframehelix.gif

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